Lenin and Stalin on Russian TV

Forecasting historical references in a television archive

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Forecasting problem

How many references to Lenin and Stalin will appear on four federal TV channels over the next 31 days?

Why should we care?

Study political narratives through historical references.
Track changes over time and compare channels.
Build a profile of editorial framing and examine possible propagandistic uses of historical figures.

Forecast specification

TARGET VARIABLES

Daily counts of mentions and archive records with at least one mention

HORIZON AND INFORMATION CUTOFF

3 Oct - 2 Nov 2026: 31 days
Information available through 2 October

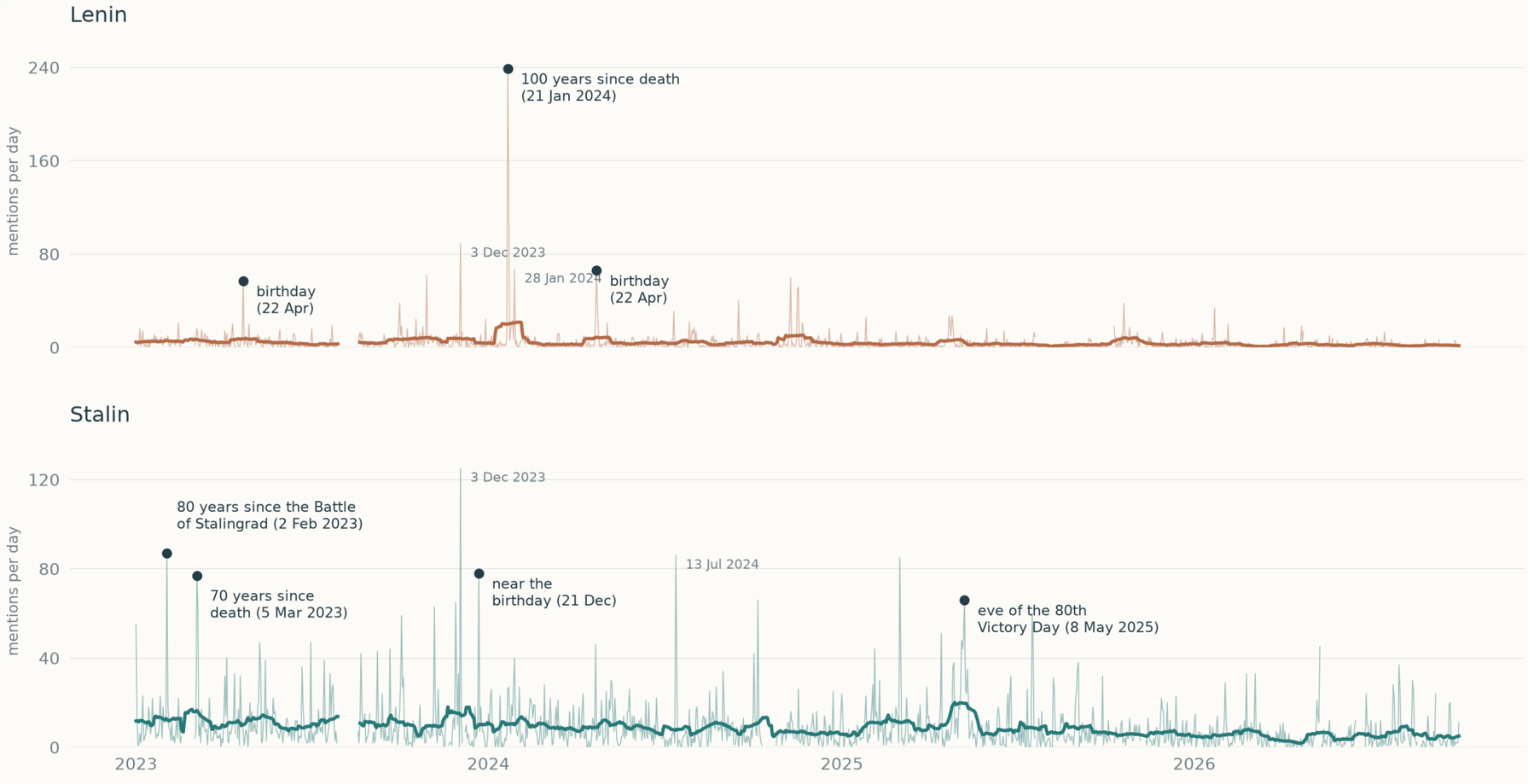
FORECAST SHAPE

Predictive distributions for integer counts; means and 80% / 95% predictive intervals

Mention counts alone do not measure news quality or establish that a message is propaganda.

Data: outcomes and forecasting information

GDELT broadcast transcripts, Jan 2023 - Oct 2026; Channel One, NTV, Russia 1 and Russia 24



Labelled: anniversaries from the model calendar; grey: the largest peaks without a calendar date. A coinciding date does not imply a cause.

Outcomes

Mentions and records with at least one mention
2 figures × 4 channels × 2 targets = 16 series

Predictor information

Past archive counts
Channel and time trend
Month and weekday
Known anniversary dates

5,334 / 5,484

observed channel-days
Archive gaps are not zeros

Names are screened by context to distinguish person references from named objects and other people.

Models and the calendar patterns they capture

M1 is the baseline; M3HR extends M3 with partial pooling and a shared calendar-month effect

Models compared

M1 — **Constant rates**
Poisson / Binomial baseline

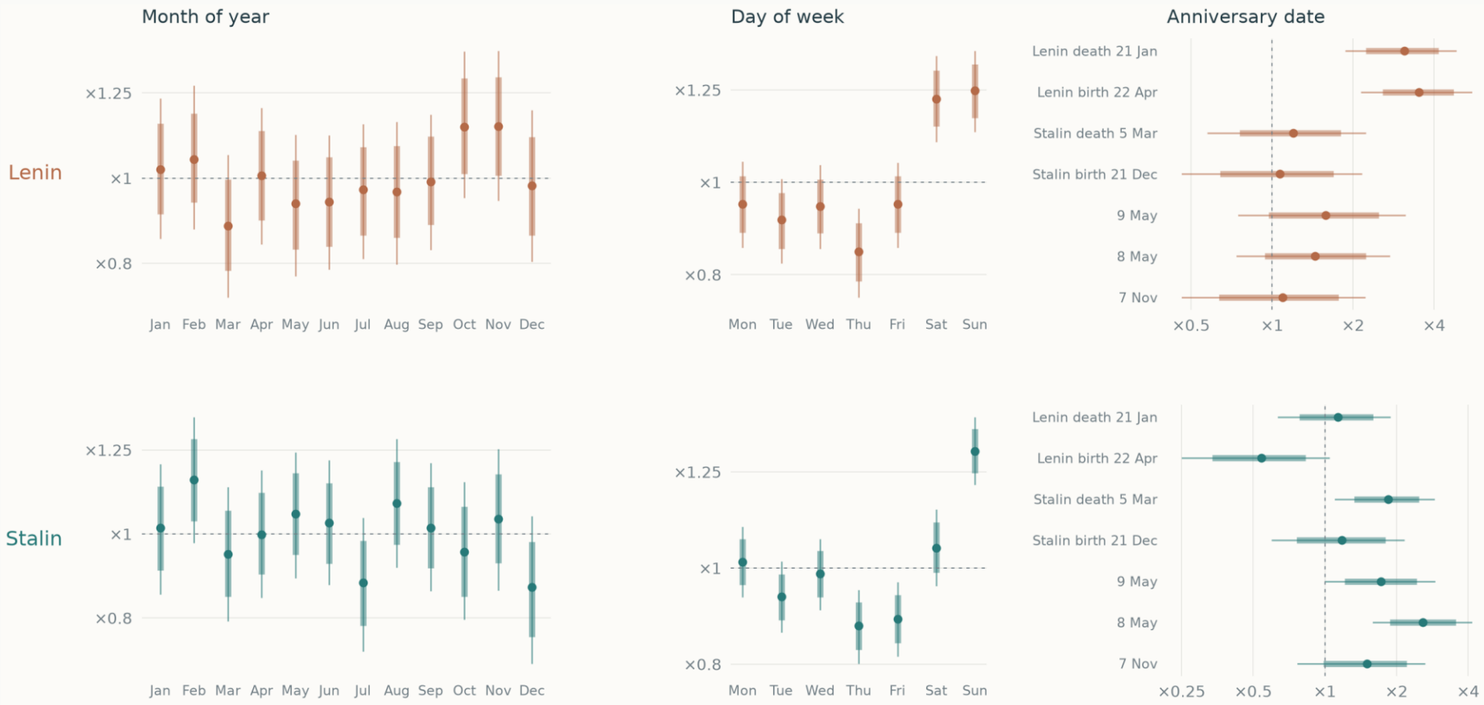
M2 — **Overdispersion**
Allows bursty daily counts

M3 — **Calendar + trend**
Month, weekday, anniversaries

M4 — **Dynamic level**
M3 + random-walk levels

M3HR — **Final model**
M3 + estimated shrinkage
+ shared month effect

Final-model posterior calendar effects



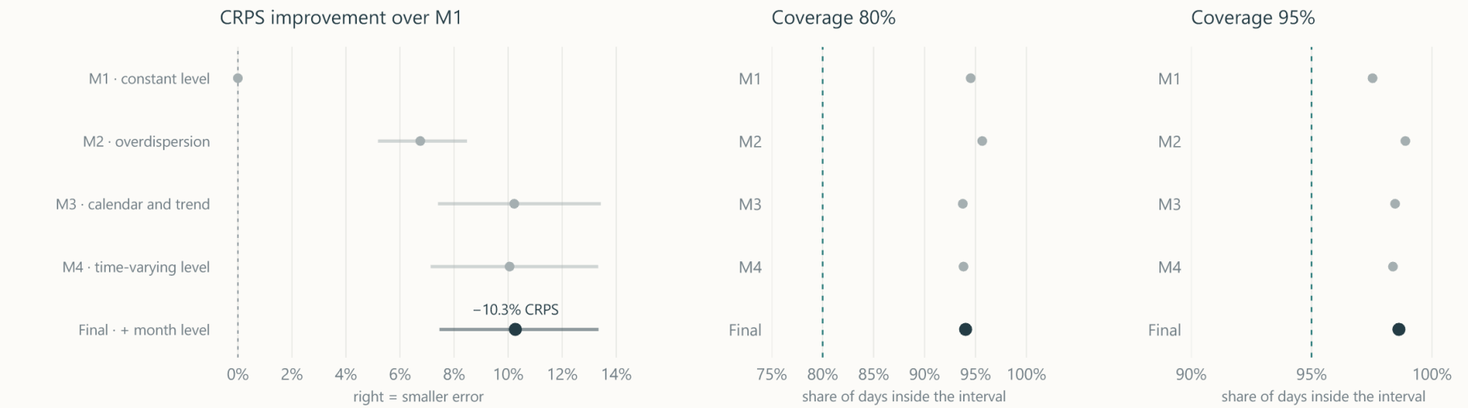
Component "a record mentions the figure" of the final model (beta-binomial, logit). Month effects are shrunk towards a common level (L4). Coefficients are conditional, not causal.

Odds multipliers for a record mentioning each figure.
1 = no effect; bands show 80% and 95% posterior intervals.

Shared structure: predict archive volume, records with a mention, then total mentions. Monthly intervals use joint trajectories.

Backtesting results: daily accuracy and monthly uncertainty

21 expanding-window test months, Jan 2025 - Sep 2026; models use the same observations

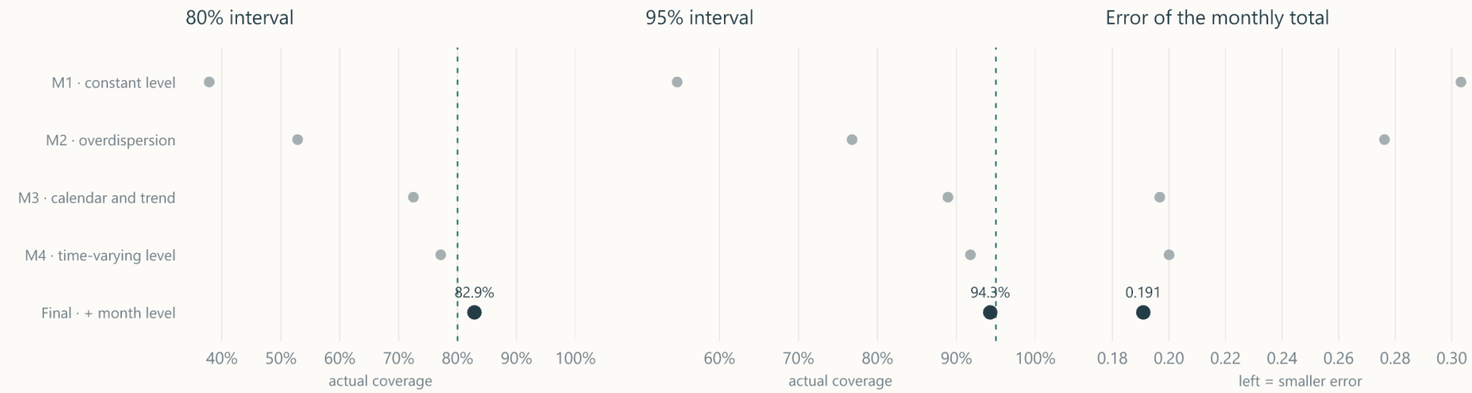


Bars on the left: 95% block bootstrap (3-month blocks).
Final = M3 + shared month level + estimated shrinkage of month effects (L4); its structure was chosen after the outer test, so the scores are optimistic.

Daily distributions

10.3% lower CRPS than baseline M1

Only 0.046% below M3. Most gain comes from calendar effects.



Dashed line - nominal level. The month level shared by channels widens monthly intervals to the nominal level.
Final = M3 + shared month level + estimated shrinkage of month effects (L4); its structure was chosen after the outer test, so the scores are optimistic.

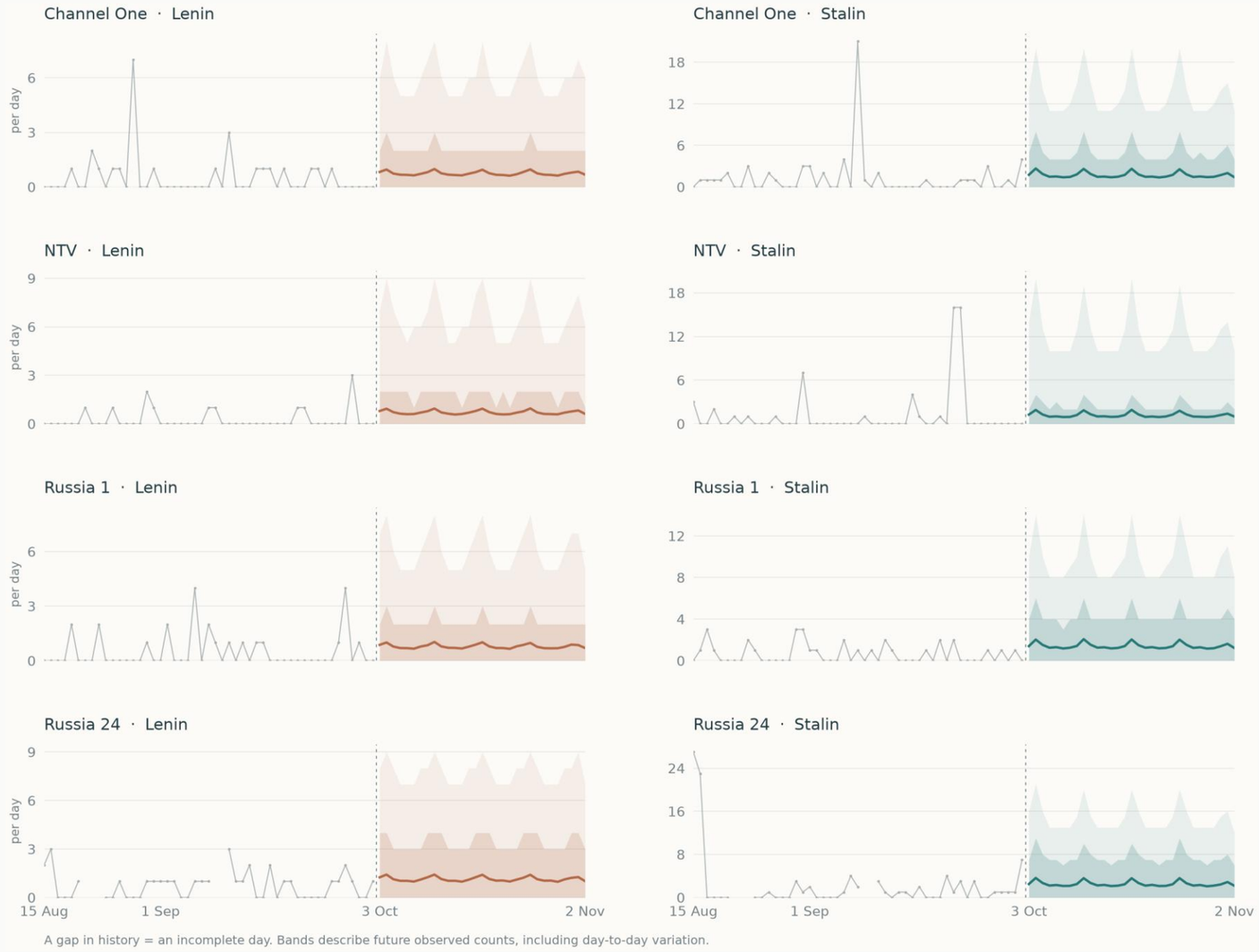
Monthly totals

2.96% lower CRPS than M3

80% coverage: 82.9%
95% coverage: 94.3%

Daily forecast: mentions on each channel

M3HR, 3 Oct - 2 Nov 2026; predictive distributions for daily mention counts



A gap in history = an incomplete day. Bands describe future observed counts, including day-to-day variation.

What is shown?

Grey: observed history
Line: predictive mean
Bands: 80% and 95% predictive intervals

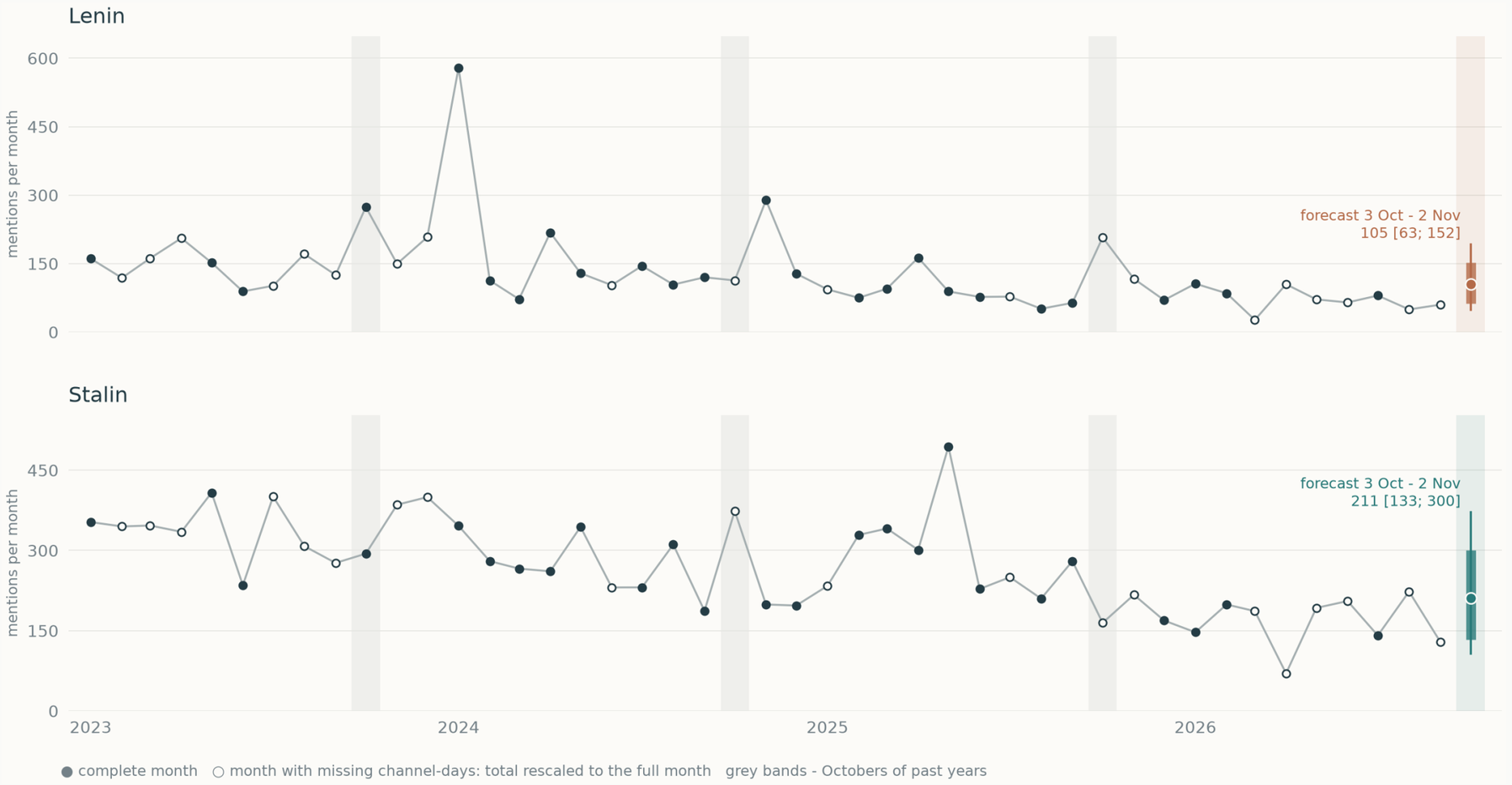
How to read it

Small daily means coexist with occasional bursts. Intervals include daily count variation and parameter uncertainty.

Conclusion: daily accuracy is stable; monthly uncertainty improves. Validate M3HR on new forecast origins.

M3HR forecast: 3 October - 2 November 2026

Mentions, sum of four channels; point: predictive mean, thick line: 80%, thin line: 95%



Lenin

105

mentions over 31 days

80%: 63-152
95%: 49-192

Stalin

211

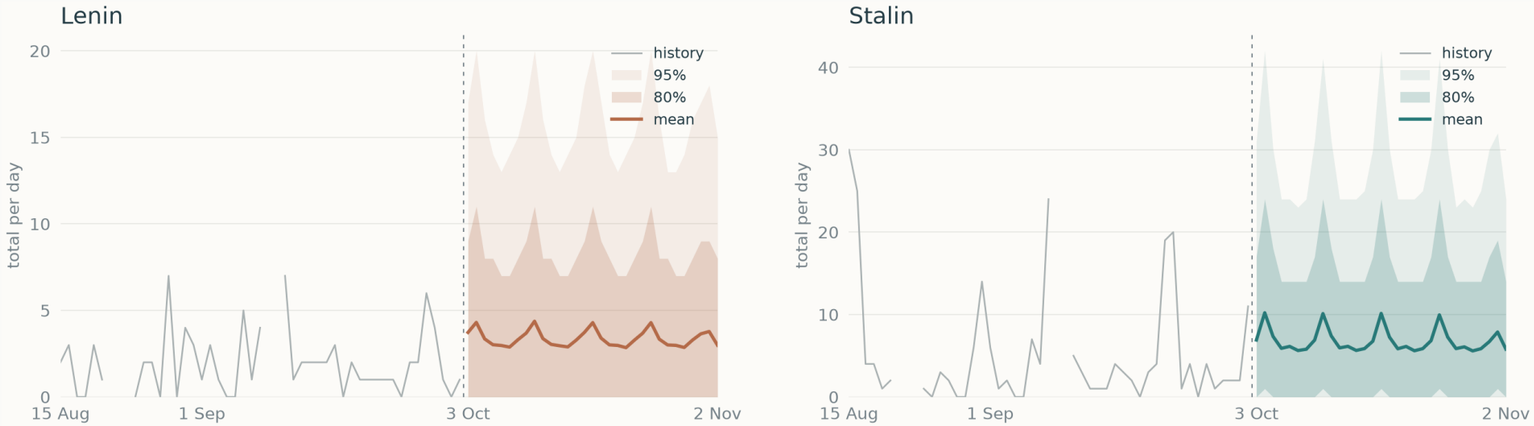
mentions over 31 days

80%: 133-300
95%: 107-371

Daily quality is largely unchanged; monthly uncertainty is better represented. Check future forecast origins.

Daily totals across all four channels

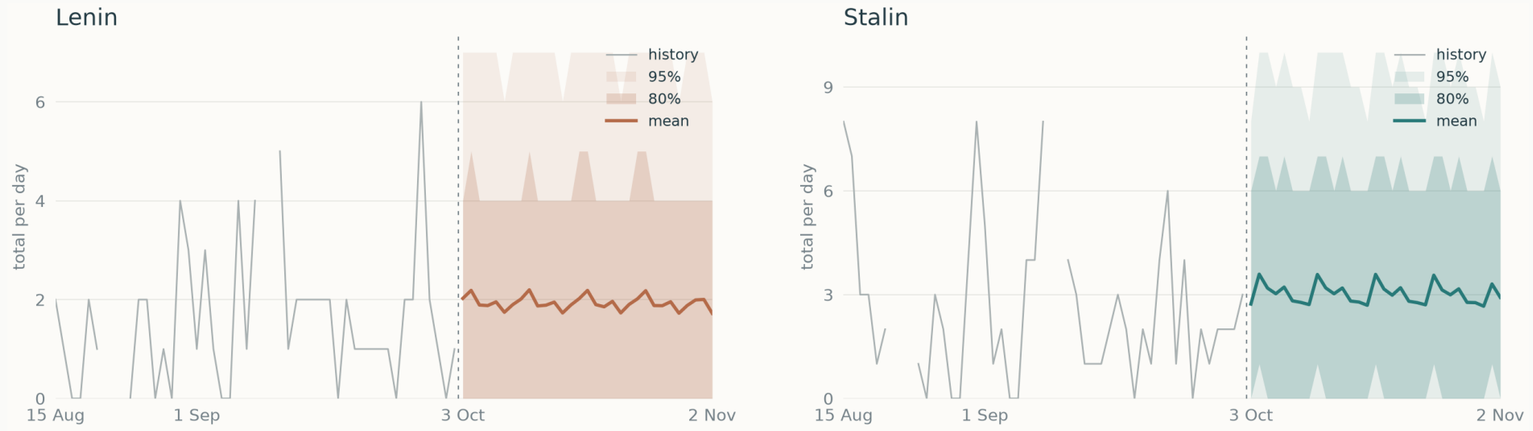
M3HR, 3 Oct - 2 Nov 2026; mean and 80% / 95% predictive intervals



Channels are summed within each joint trajectory before computing 80% and 95% intervals. A gap = an incomplete archive day.

All mentions

Daily sum across channels.
Left: Lenin
Right: Stalin



Channels are summed within each joint trajectory before computing 80% and 95% intervals. A gap = an incomplete archive day.

Records with a mention

The second target counts each archive record once, even with repeated mentions.

Channels are summed within each joint trajectory before calculating predictive intervals.